

Overlapping Generations in Lean: The Life-Cycle Household

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Abstract

This is the record of a programme to formalise overlapping-generations economies in Lean 4, and of its first stage: the life-cycle household. A household that lives J periods against uninsurable earnings risk and a borrowing constraint solves a finite backward induction, and everything proved for the infinite-horizon Bellman operator with an arbitrary continuation applies stage by stage with no fixed point. Beyond that we prove the finite-horizon sandwich: with k periods still to come, consumption lies between $\kappa_k m$ and $\kappa_k(m + H_k(z))$, where $1/\kappa_k = \sum_{i \leq k} (p/R)^i$ is the consumption share of a household with no future income and H_k is the expected present value of the next k incomes; the propensities decrease in the horizon towards the infinite-horizon $\kappa = 1 - p/R$. We then add retirement and a pension. With a lump-sum pension the retired phase is an income-fluctuation problem in its own right and everything survives. With a means-tested pension we settle a claim often made informally: a means test does make the policy non-monotone, but not for the reason usually given. Optimal saving rises with cash on hand whatever the continuation, by Topkis, so a value function made non-concave by the test cannot by itself reverse it; what the test reverses is consumption, which jumps down at the wealth where the household gives the pension up, and cash on hand itself, once the taper is steeper than the gross return. Finally we report what the life cycle does for Light's Theorem 1, that saving rises with the interest rate: it holds at every age, exactly, below a critical degree of relative risk aversion of about 3.4 at Aiyagari's calibration, and fails above it, so the life cycle delivers his $\mu = 3$ and not his $\mu = 5$. The programme's target is uniqueness of the stationary equilibrium at relative risk aversion above one, which the infinite horizon could not close: in a life cycle every comparative static is a finite induction, wealth is bounded, and the age-one distribution is common to every rate.

JEL: C63, D52, E21. *Keywords:* overlapping generations, life cycle, Bewley models, formal verification, Lean.

*The Lean development described here is at <https://github.com/LeanEconomics/LeanEconomics>. It was written mostly by Claude (Anthropic), an AI coding model, working under the direction of Robert Kirkby (Victoria University of Wellington); every result reported was checked by the Lean kernel. This is the companion to the write-up on Aiyagari (1994) in the same repository. Comments welcome.

1 Introduction

The companion write-up in this repository proved that the stationary equilibrium of Aiyagari (1994) is unique at his logarithmic calibration, and located exactly what stands in the way for relative risk aversion $\mu \in \{3, 5\}$. Every obstacle it met traces to the fixed point of the infinite-horizon problem: a condition on the response of consumption to the interest rate fails in a tail of wealth that the stationary distribution reaches; an induction that would prove the response bound has a step multiplier $1/\mathfrak{p} > 1$, so it cannot close as a supremum argument; and the only floor on capital supply comes from an ergodic identity that loses a factor $\|u'\|^2/\text{Var}$. Overlapping generations remove all three at once. A household that lives J periods solves a finite backward induction from an explicit terminal condition, so a multiplier $1/\mathfrak{p}$ per step compounds to $\mathfrak{p}^{-J}$, which at Aiyagari's numbers is about 1.03. Wealth is bounded by J periods of accumulation, so any condition needs checking on a known bounded region. And the distribution of the youngest cohort is common to every interest rate, so an ordering of policies pushes forward to an ordering of the whole age profile of distributions by induction on age, with no Doeblin atom and no uniqueness of a stationary distribution.

The target of the programme is uniqueness of the stationary equilibrium of a stochastic life-cycle economy in the manner of Huggett (1996) at $\mu \in \{3, 5\}$. Section 2 records the first stage, the life-cycle household, which is formalised. Section 6 records the rest of the programme, the classical results of Diamond (1965) included, and what each stage needs. As in the companion, every result stated with a formal name has been checked by the Lean kernel against Mathlib; the development is at <https://github.com/LeanEconomics/LeanEconomics>, directory `LeanEconomics/OLG`.

2 The life-cycle household

2.1 Stages

The household of the companion write-up is kept: earnings $y(z)$ follow a finite Markov chain π , the gross return is $R = 1 + r$, cash on hand is $m = Ra + y(z)$, saving $a' \in [0, \bar{a}]$ is bounded below by the borrowing limit and above by an artefactual cap, utility is CRRA with $u'(c) = c^{-\mu}$ and $\beta \in (0, 1)$. What changes is the horizon: the household lives J periods and leaves nothing behind. Count periods from the end. With k periods still to come after today, the continuation value is the k -th Bellman iterate from zero,

$$V_k = T^k 0, \quad (Tv)(a, z) = \max_{a'} \{u(m - a') + \beta \sum_{z'} \pi(z, z') v(a', z')\},$$

and the household with k periods to come saves $g_k = \text{policyOf } V_k$ and consumes $c_k = m - g_k$ (`stageValue`, `stagePolicy`, `stageConsumption`). Age j of a J -period life is stage $k = J - 1 - j$. In the last period the household saves nothing and consumes its cash on hand (`stagePolicy_zero`, `stageConsumption_zero`).

This is the finite-horizon problem with a flat age profile of earnings: the same income

process at every age. An age profile, retirement and survival risk need income to be a parameter of the Bellman step rather than a field of the structure, and are deferred to the next stage of the programme.

2.2 What transfers without a fixed point

The companion development proved its household results for the Bellman operator applied to an arbitrary continuation v , and only then passed to the fixed point. Here there is no fixed point to pass to, and each result is read at $v = V_k$:

- consumption is positive at every stage when marginal utility is unbounded at zero (`stageConsumption_pos`);
- the stage value has concave slices, so the optimal action is unique and both saving and consumption rise with wealth (`stagePolicy_mono`, `stageConsumption_mono`);
- Carroll–Kimball: consumption is concave in wealth at every stage (`concaveOn_stageConsumption`);
- the Euler inequality under the cap, $\beta R \mathbb{E}[u'(c_k(a', z')) \mid z] \leq u'(c_{k+1}(a, z))$, at every state (`stage_euler_le`), and the Euler inequality above the floor, the reverse, at every state where the household saves (`stage_euler_ge`).

2.3 The finite-horizon sandwich

Let $\mathfrak{p} = (\beta R)^{1/\mu}$ and define the finite-horizon minimal propensities to consume by

$$\kappa_0 = 1, \quad \kappa_{k+1} = \frac{\kappa_k R}{\mathfrak{p} + \kappa_k R}, \quad \text{so that} \quad \frac{1}{\kappa_k} = \sum_{i=0}^k \left(\frac{\mathfrak{p}}{R} \right)^i$$

(`stageMPC`): the share of cash on hand consumed by a household with k periods to come and no future income. Let $H_0 = 0$ and $RH_{k+1}(z) = \sum_{z'} \pi(z, z')(y(z') + H_k(z'))$, the expected present value of the next k incomes (`stageHumanWealth`).

Theorem 2.1 (The finite-horizon sandwich). *Assume CRRA utility with marginal utility unbounded at zero, and that the cap is slack at every stage. Then for every k , every z and every $a \in [0, \bar{a}]$,*

$$\kappa_k m \leq c_k(a, z) \leq \kappa_k (m + H_k(z)).$$

Formal statement: `IncomeFluctuation.stageMPC_mul_le_stageConsumption`,
`IncomeFluctuation.stageConsumption_le_stageHumanWealth`.

Each bound is one Euler step, applied k times. The lower bound: if consumption against v is at least κm , the Euler inequality above the floor at a state where the household saves gives $c^{-\mu} \leq \beta R \mathbb{E}[(\kappa R a')^{-\mu}]$ with $a' = m - c$, hence $\kappa R(m - c) \leq \mathfrak{p}c$ and $c \geq \kappa R(\mathfrak{p} + \kappa R)^{-1}m$; at a corner $c = m$ (`crra_stageMPC_step`). The upper bound: if consumption against v is at most $\kappa(m + H(z))$, the Euler inequality under the cap and Jensen for the negative power give

$c^{-\mu} \geq \beta R (\mathbb{E}[c'])^{-\mu}$ with $\mathbb{E}[c'] \leq \kappa R(a' + H'(z))$, hence $c \leq \kappa R(\mathfrak{p} + \kappa R)^{-1}(m + H'(z))$ (`ccra_stageUpper_step`). The recursion for κ is the same on both sides, which is why the sandwich is exact in the homogeneous case.

The propensities decrease in the horizon towards the infinite-horizon $\kappa = 1 - \mathfrak{p}/R$ of the companion write-up: $\kappa \leq \kappa_k$ for every k (`minMPC_le_stageMPC`), so a household with less life left consumes a larger share of its cash, and the infinite-horizon bounds are the limiting case. A first consequence for the programme is an explicit bound on saving, $g_k(a, z) \leq (1 - \kappa_k)(Ra + y_{\max})$ (`stagePolicy_le`), which iterated from zero wealth at the start of life bounds the wealth of every cohort.

3 Retirement, a pension, and the means test

3.1 Retirement with a lump-sum pension

The household works until age J_r and is retired after it, receiving a pension in place of its labour earnings. With a *lump-sum* pension p the retired household faces the same problem as the working one with income replaced by the constant p , so the retired phase is itself an income-fluctuation problem (`withPension`) and everything in Section 2 applies to it unchanged. The life-cycle value is assembled by iterating the working Bellman operator on the retired value: `lifeValue( $n_r, n_w$ )` is the value of a household with n_w working and n_r retired periods still to come (`lifeValue`), so a household of age $j < J_r$ in a life of J periods sits at $n_w = J_r - j$ and $n_r = J - J_r$. Saving and consumption rise with wealth at every age (`lifePolicy_mono`, `lifeConsumption_mono`).

Two facts about the pension itself are worth recording. A retiree's human wealth is the same in every earnings state, because the pension does not depend on the earnings shock (`stageHumanWealth_withPension_const`), and it never exceeds the perpetuity value of the pension, $H_k \leq p/r$ (`stageHumanWealth_withPension_le`). With Theorem 2.1 the retiree's consumption therefore satisfies

$$\kappa_k m \leq c_k(a) \leq \kappa_k \left(m + \frac{p}{r} \right)$$

at every retired age. And cash on hand is strictly increasing in assets (`withPension_resources_strictMono`). Both of these fail once the pension is means-tested.

3.2 Means testing

A pension is means-tested when the amount paid falls with the recipient's assets. We write the schedule as a function $p(\cdot)$ of assets and cash on hand as $m(a) = Ra + p(a)$ (`cash`), and consider three schedules (`lumpSum`, `tapered`, `cliff`): a lump sum; a taper, withdrawn at rate τ per unit of assets above a threshold and never negative; and a cliff, withdrawn entirely above the threshold.

It is often argued that a means test makes the household's policy function non-monotone, and the reason usually given is that the test makes the value function non-concave. The

argument as usually given is wrong, and the conclusion is nevertheless true, for a different reason. Three results separate the two.

Theorem 3.1 (Monotonicity in cash on hand is never lost). *Let period utility be strictly concave on the positive half-line and let W be any continuation value. If $m_1 < m_2$ and x_i is optimal at m_i , then $x_1 \leq x_2$.*

Formal statement: *OLG.isOptimalSaving_mono.*

The proof is Topkis. Adding the two optimality inequalities gives $u(m_1 - x_1) + u(m_2 - x_2) \geq u(m_1 - x_2) + u(m_2 - x_1)$; if $x_2 < x_1$ the two right-hand consumptions are strict convex combinations of the two left-hand ones with the same sum, so strict concavity makes the reverse inequality strict, a contradiction. The continuation enters only through terms that cancel. So a non-concave value function — which a means test certainly produces — cannot by itself make saving fall with cash on hand, whatever the shape of W .

What a non-concave continuation does produce is a jump, and the jump falls on consumption.

Theorem 3.2 (Consumption falls when wealth rises). *There are a continuation value arising from a means-tested pension and two levels of cash on hand, the second larger, at which optimal consumption is strictly smaller. With log utility, a unit gross return, and a pension of one withdrawn entirely above assets of one: at $m = 6$ the household saves exactly 1 and consumes 5; at $m = 8$ it saves 4 and consumes 4.*

Formal statement: *OLG.exists_consumption_decrease.*

At $m = 6$ the household keeps the pension, saving the most it can without losing it; at $m = 8$ the pension is no longer worth keeping, so the household gives it up and saves the proceeds. Saving rises from 1 to 4, as Theorem 3.1 says it must, and since it rises by more than wealth does, consumption falls. Both optima are exact: the objective is log of a quadratic on each side of the threshold, and the comparison is checked by the kernel.

The third result is the genuine non-monotonicity of the saving policy, and it comes from the budget rather than from the value function. A retiree's pension is assessed each period on the assets then held, so the schedule enters cash on hand directly.

Theorem 3.3 (A gentle taper is harmless; a cliff is not). *If $0 \leq \tau < R$ then $a \mapsto Ra + p(a)$ is strictly increasing under the taper (*cash_strictMono_tapered*), and so, by Theorem 3.1, saving is monotone in assets (*isOptimalSaving_mono_assets*). Under a cliff it is not: if $a \leq \bar{a} < b$ and $R(b - a) < p$ then cash on hand is strictly smaller at b (*cash_lt_cash_of_cliff*), and there are two asset levels, the second larger, at which the household saves strictly less: with a concave continuation, at assets 1 it saves $1/2$ and at assets $3/2$ only $1/4$.*

Formal statement: *OLG.exists_saving_decrease_in_assets.*

The continuation in the second half of Theorem 3.3 is $\log(x + 1)$, which is concave: nothing is hidden in it. The non-monotonicity is entirely the budget's, and the threshold is sharp, τ against R . A taper gentler than the gross return leaves the household's saving monotone in assets no matter how badly the test distorts the value function; a taper steeper than the gross return, or a cliff, makes a richer household poorer in the only sense that matters for today's decision.

3.3 Which policy, and in which variable

The two non-monotonicity results are about different objects, and it is worth separating them in the household's own state space, assets a and the Markov income shock z .

Next period's assets are monotone. For each z , the optimal a' is nondecreasing in a whatever the continuation value, with no concavity assumed of it:

Theorem 3.4 (a' rises with a , at each z). *Let $a < a'$ both lie in the asset region. Then for every continuation v and every z , $g_v(a, z) \leq g_v(a', z)$.*

Formal statement: *IncomeFluctuation.policyOf_mono_of_lt.*

This is Theorem 3.1 read in the state space: cash on hand rises with assets, so optimal saving does. The repository already had this statement (`policyOf_mono`) but with concave slices of the continuation as a hypothesis, which is exactly what a means test destroys; the Topkis argument removes the hypothesis, at the cost of requiring the two asset levels to be distinct. The continuation cancels between the two optimality inequalities and only strict concavity of period utility survives.

Consumption is not monotone, and the contrast is sharp rather than an artefact of proof technique. Consumption is $m(a) - g_v(a, z)$, and both terms rise with a ; when the continuation is non-concave the second can rise faster, which is exactly Theorem 3.2. So monotone consumption genuinely requires concavity of the continuation, while monotone saving does not.

Two things follow for the means test. Where the pension enters only next period's problem, so that today's cash on hand is unaffected, Theorem 3.4 applies verbatim: next period's assets still rise with this period's, and only consumption misbehaves. Where the test is assessed on the assets currently held, the hypothesis of Theorem 3.4 is the one that fails: cash on hand is no longer increasing in a , and with it the policy. That is the whole of Theorem 3.3, and the dividing line is τ against R .

3.4 At realistic parameters

The three theorems predict exactly which of the two channels operates for which schedule, and a calibrated retiree confirms it. We solve a deterministic retiree of 25 years by backward induction with global search over a grid of 6001 asset levels — the value function is not concave, so a first-order condition cannot be used — at $r = 4\%$, $\beta = 0.96$, relative risk aversion 3, a full pension of 0.40 of mean earnings and a free area of 3 mean-earnings units (`numerics/meanstest.m`). The taper of 0.078 per unit of assets is Australia's asset test, three dollars a fortnight per thousand dollars.

Pension schedule	Cash on hand	Saving	Consumption
Lump sum	increasing	increasing	increasing
Taper 0.078 (Australia)	increasing	increasing	falls by 0.025–0.046
Taper 0.50	increasing	increasing	falls by 0.13–0.24
Taper 1.20 (above R)	falls	falls by 0.004	falls by 0.55–0.70
Cliff	falls by 0.40	falls by 0.32–0.37	falls by 0.55–0.70

Every entry is what the theorems require. Saving falls with assets in exactly the two rows where cash on hand does, which are exactly the two rows whose taper exceeds the gross return, as Theorem 3.3 says; in the other three it rises, as Theorem 3.1 says it must. Consumption falls in every row with a test of any kind, as Theorem 3.2 says it can, and the drop sits exactly at the free area and grows with age, from 0.025 of mean earnings at the start of retirement to 0.046 twenty years in under the Australian taper. The mechanism is the one the witness isolates: just above the free area the effective return to saving is only $R - \tau$ until the pension is exhausted, so the household jumps its saving toward the far side of the taper range and cuts consumption to pay for it.

The effect is not confined to retirees. A worker in the last year before retirement, facing the Australian taper for the whole of retirement, has a monotone saving policy and a consumption policy that falls by 0.025 of mean earnings at the same threshold: the non-concavity is inherited through the continuation value, and it is again consumption that carries it.

4 Stationary equilibrium

Every cohort is born with no assets and an earnings state drawn from a fixed law ν , lives J periods, and is replaced; cohorts have equal weight. The distribution of the youngest cohort is therefore the same at every interest rate, and the distribution of every older cohort is that law pushed forward by the policies of the ages in between. This is the structural advantage of the life cycle over the infinite horizon, and it removes most of the machinery the stationary case needed.

4.1 The aggregate is a finite composition

Because assets at birth are degenerate at zero, no measure theory is required. Let

$$(\mathcal{S}_k h)(a, z) = \sum_{z'} \pi(z, z') h(g_k(a, z), z')$$

be the expectation of h next period for a household with k periods still to come (`stageStep`). The assets a household holds over the rest of its life, summed across ages and seen from its current state, satisfy

$$A_0(a, z) = a, \quad A_{k+1}(a, z) = a + (\mathcal{S}_{k+1} A_k)(a, z)$$

(`cohortAssets`), and aggregate capital per head is $K = \frac{1}{J} \sum_z \nu(z) A_{J-1}(0, z)$ (`olgCapital`). There is no pushforward measure, no Feller property, no stationary distribution, no Doeblin condition and no uniqueness of an invariant law: the equilibrium object is a finite sum of finite sums, and the recursion above is its definition.

4.2 The chain

Theorem 4.1 (Light’s Theorem 2 for the life cycle). *Suppose two economies share the earnings chain and their policies are ordered at every age, $g_k^{(1)} \leq g_k^{(2)}$ on the asset region. Then $A_k^{(1)} \leq A_k^{(2)}$ for every k , and aggregate capital is ordered, $K^{(1)} \leq K^{(2)}$.*

Formal statement: *IncomeFluctuation.cohortAssets_le, IncomeFluctuation.olgCapital_le_of_stagePolicy_le.*

The proof is an induction on age with two ingredients, and both are already available. The class of functions increasing in assets is preserved by $\mathcal{S}_k$ (`monoRegion_stageStep`), because next period’s assets rise with this period’s at each earnings state and for any continuation, which is Theorem 3.4. And ordered policies keep ordered functions ordered (`stageStep_le_stageStep`): the higher-rate economy evaluates a larger function at a larger asset level. Note what is *not* needed. The infinite-horizon version of this step required monotone transitions, a coupling argument and first-order stochastic dominance of the stationary distributions; here the whole statement is an inequality between two functions on the same finite state space, because the distribution the two economies start from is literally the same.

4.3 A floor on saving from primitives

Existence needs capital supply at the top of the rate interval to exceed demand. In the infinite horizon the only floor available came from an ergodic identity that loses a factor $\|u'\|_\infty^2 / \text{Var}$, with the consequences for the constants reported in the companion write-up. The life cycle needs nothing of the kind: the upper sandwich of Theorem 2.1 bounds consumption at every age, and what is not consumed is saved.

Proposition 4.2 (Affine floor on saving). *At every age and every state in the region,*

$$g_k(a, z) \geq (1 - \kappa_k)(y_{\min} + Ra) - \kappa_k H_k(z).$$

Formal statement: *IncomeFluctuation.le_stagePolicy.*

Iterated from zero assets at birth this is an explicit floor on the assets of every cohort, in the primitives alone: the discount factor, the rate, the earnings bounds and the horizon.

4.4 Uniqueness

Theorem 4.3 (Single crossing). *A capital supply that never falls with the rate meets a strictly falling demand at most once.*

Formal statement: *IncomeFluctuation.olgEquilibriumRate_unique.*

With Theorem 4.1 this reduces uniqueness of the stationary equilibrium to Light’s Theorem 1, that saving rises with the rate at every age, which is the subject of Section 5: it holds at Aiyagari’s calibration for relative risk aversion up to about 3.4, so the chain closes at his $\mu = 3$ and not at his $\mu = 5$.

4.5 Continuity, and existence

The last input is continuity of capital supply in the rate. In the infinite horizon this was a substantial development, because the stationary distribution itself had to be shown to move continuously. Here there is no stationary distribution: capital supply is a finite composition of finitely many policies, so all that is needed is that each policy moves continuously with the rate, and that continuity composes.

We take the route of the companion write-up and put the interest rate into the state, so that Berge’s theorem gives continuity in assets and the rate jointly rather than continuity in a parameter. That device was used there at the fixed point, but everything it rests on — that the feasible sets and the objectives agree on the slice — holds for an arbitrary continuation. So it works stage by stage, and here it is easier, because the slice of the augmented Bellman iterate is the iterate of the sliced programme by a plain induction, with no fixed point to identify (`sliceAt_bellman`, `sliceAt_augStageValue`). The augmented argmax at each stage is then a singleton, since the sliced one is by Carroll–Kimball concavity of the iterates (`argmax_aug_stage_eq_singleton`), so the stage policy is jointly continuous (`continuousOn_augStagePolicy`) and agrees with the sliced one (`augStagePolicy_eq`). Running the cohort recursion in the augmented state (`augCohortAssets`) gives a composition of continuous maps that reads, at a rate in range, as capital supply at that rate (`augCohortAssets_eq`, `olgSupply_eq`).

Theorem 4.4 (Capital supply is continuous). *$r \mapsto K(r)$ is continuous on $[r_{lo}, r_{hi}]$.*

Formal statement: *`IncomeFluctuation.continuousOn_olgSupply`.*

Theorem 4.5 (Existence). *If capital supply lies below demand at the bottom of the interval and above it at the top, and demand is continuous, then the two meet: there is a rate in $[r_{lo}, r_{hi}]$ at which the capital the cohorts accumulate equals the capital the firm demands.*

Formal statement: *`IncomeFluctuation.exists_olgEquilibrium`.*

With Theorem 4.1 and Theorem 4.3 the life-cycle economy therefore has the same three-part structure as the stationary one — existence, monotone supply, single crossing — with every piece proved and only Light’s Theorem 1 assumed. The two inequalities Theorem 4.5 needs are the ceiling at the bottom of the interval, which the budget gives, and the floor at the top, which Proposition 4.2 gives in the primitives, with none of the constants that the ergodic route forced on the stationary case.

5 Theorem 1 in the life cycle: how far it goes

The programme of Section 6 rests on Light’s Theorem 1, that saving rises with the interest rate at every state. In the infinite horizon this is open above logarithmic utility. The life cycle was the reason for hope: a finite backward induction from an explicit terminal condition, with the per-step loss compounding only J times instead of to a fixed point. This section reports what the numbers say, at $\beta = 0.96$, a seven-state Tauchen chain with $\rho = 0.9$ and $\sigma = 0.4$, a zero borrowing limit, $J = 60$, and a flat earnings profile (`numerics/lc_check.m`, `lc_robust.m`, `lc_induct.m`, `lc_verify.m`, `lc_bound.m`, `lc_crit.m`).

5.1 The statement is its own terminal condition

The budget constraint makes two statements identical. Since $a' = Ra + y(z) - c$ at either rate,

$$g_k(a, z; r_2) \geq g_k(a, z; r_1) \iff c_k(a, z; r_2) - c_k(a, z; r_1) \leq (r_2 - r_1)a,$$

so Theorem 1 is the affine bound on the rate response of consumption with slope one and zero intercept. That bound holds with equality in the last period of life, where $c_0 = y(z) + Ra$. The question is therefore whether the terminal condition propagates backward, and it is a question about one inequality rather than about a pair of constants to be chosen.

5.2 It propagates, up to a critical risk aversion

At $\mu = 1$ and $\mu = 3$ it propagates exactly. Over all sixty stages and the whole wealth region reachable from zero wealth at birth, the largest violation of $g_k^{(1)} \leq g_k^{(2)}$ is zero, and the consumption bound holds to five parts in 10^{14} ; refining the asset grid from 2000 to 16000 points changes nothing. At $\mu = 5$ it fails. The failure is a genuine feature of the household's problem, not of the solution method: the violation is identical at every grid size, and endogenous-grid and value-function iteration with a pure grid search agree on it to the resolution of the grid. Saving at zero wealth in the highest income state, in the first period of a sixty-period life, falls as the rate rises:

r	2%	3%	4%	5%	6%
$g(0, z_{\max})$, endogenous grid	1.7783	1.7638	1.7452	1.7234	1.6993
$g(0, z_{\max})$, value iteration	1.7806	1.7606	1.7406	1.7206	1.7006

Between the two lies a critical degree of risk aversion, and it is sharp. Bisecting on μ for the largest violation over all stages gives:

Horizon J	$r = 2\%$	$r = 4\%$	$r = \lambda = 4.17\%$
20	3.79	3.53	3.51
40	3.77	3.39	3.36
60	3.77	3.39	3.36

Theorem 1 holds throughout the life cycle below the critical value and fails above it. The value is stable in the horizon beyond about forty periods, and falls as the rate rises towards the rate of time preference. Aiyagari's $\mu = 3$ is inside it with little room, at $\mu = 3.39$ against 3; his $\mu = 5$ is well outside. So the life cycle buys one of the two open cases and not the other, and it buys $\mu = 3$ only at rates below the rate of time preference, which is where the equilibrium lies.

5.3 Why the natural induction does not close

The backward induction that the transfer lemma suggests compares the two Euler equations at a state where the lower-rate household saves. Writing $b = g_{k+1}^{(1)}(a, z)$ for the pivot, the

induction hypothesis and monotonicity of c_k give $c_k^{(2)}(b') \leq c_k^{(1)}(b) + \Delta r b$ at the relevant points, and the step closes provided

$$R_2 \mathbb{E}[(c_k^{(1)}(b) + \Delta r b)^{-\mu}] \geq R_1 \mathbb{E}[c_k^{(1)}(b)^{-\mu}] \left(1 + \frac{\Delta r a}{c_{k+1}^{(1)}(a, z)}\right)^{-\mu}.$$

To first order this asks $\mu(b/c_k^{(1)}(b) - a/c_{k+1}^{(1)}(a, z)) \leq 1/R$: the ratio of assets to consumption may rise from one age to the next by at most $1/(\mu R)$. It does not. At zero wealth in the highest income state the left-hand side is large, because such a household saves a great deal against a certain fall in income, while the right-hand side is about one. The margin is negative from the third stage of life onward for every degree of risk aversion we tried, logarithmic utility included, at -0.12 for $\mu = 1$, -2.86 for $\mu = 3$ and -6.29 for $\mu = 5$ per unit of the rate gap. Keeping the term the step discards, the marginal propensity to consume times the size of the hypothetical violation, bounds the violation by a multiple of Δr rather than eliminating it, which is not enough: the aggregate difference the equilibrium argument must sign is itself of order Δr .

5.4 The reduction, formalised, and its verdict

The backward induction above is proved in Lean, in the form the life cycle makes natural. Because the budget constraint makes the two statements identical (`stagePolicy_le_iff_consumption_le`), the terminal stage needs no hypothesis at all: in the last period both households consume their cash on hand and save nothing (`stagePolicy_mono_zero`). Corners need none either: if the lower-rate household saves nothing there is nothing to violate. What the step needs is one inequality per age and state, which we call the *age-indexed slack condition*: writing b for the lower-rate household's saving at that state and $C'(z')$ for its consumption next period,

$$R_1 \sum_{z'} \pi(z, z') C'(z')^{-\mu} \leq R_2 \sum_{z'} \pi(z, z') (C'(z') + \Delta r b)^{-\mu} \left(1 + \frac{\Delta r a}{c}\right)^\mu \quad (1)$$

(`StageSlack`).

Theorem 5.1 (The reduction). *If (1) holds at every age and every state in the region, then saving rises with the rate at every age, and aggregate capital rises with the rate.*

Formal statement: `IncomeFluctuation.stagePolicy_mono_step`,

`IncomeFluctuation.stagePolicy_mono_of_stageSlack`, `IncomeFluctuation.olgCapital_le_of_stageSlack`.

With Theorem 4.3 this closes the chain: (1) at every age gives a unique equilibrium rate. The question is therefore whether (1) is true, and the answer is the one reported above, now stated sharply. Condition (1) is exactly the margin tabulated in the previous subsection. At $\mu = 3$ it holds for the last three or four periods of life, with margins $+0.68$, $+0.42$ and $+0.18$ per unit of the rate gap, and fails from about the fifth-to-last, at -0.23 , deteriorating to -2.86 at birth. At $\mu = 1$ it holds for most of the life and still fails for the youngest ages, at -0.12 at birth.

That last observation is the important one. The condition fails even at logarithmic utility, where Theorem 1 is certainly true. So the failure is not a symptom of high risk aversion: the pointwise Euler comparison is simply the wrong instrument in a life cycle, and no sharpening of it that we could find repairs the young ages. Keeping the term the step discards, the marginal propensity to consume times the size of the hypothetical violation, only bounds the violation by a multiple of Δr , and both refinements we tried — an upper bound on the propensity to consume from concavity and the sandwich, and a self-improving bootstrap on the size of the violation — converge to a positive bound rather than to zero.

So $\mu = 3$ is not delivered by this route, and $\mu = 1$ is not either, although the companion write-up proves the latter by the rescaling argument, which is an induction on Bellman iterates and so carries over to the life cycle unchanged. What the reduction does deliver is the end of life, and the exact inequality that any proof must beat.

So the state of the programme is this. Theorem 1 is true in the life cycle for $\mu = 3$, and the statement to prove has the cleanest possible form, its own terminal condition propagated; the reduction to one inequality per age is proved; and that inequality is false for the young, at every degree of risk aversion. What is missing is an argument at zero wealth that does not go through a pointwise comparison of the two Euler equations. The same zero-wealth state is where the infinite-horizon slack condition was tightest, which suggests the two problems share a single missing ingredient rather than two.

5.5 It is not precautionary saving

A household at zero wealth in the highest earnings state saves heavily, and the obvious diagnosis is precautionary: it is insuring against the fall in earnings it expects. If that were the mechanism, precautionary-saving theory would be the place to look for the missing argument. It is not the mechanism. The same reversal happens in the *certainty-equivalent* household, which faces the conditional mean earnings path and no risk at all, and there everything is in closed form (`numerics/precaut2.m`, `durcheck.m`).

d(saving)/dr at zero wealth	$\mu = 1$	$\mu = 3$	$\mu = 5$	$\mu = 8$
Certainty equivalent (mean path, no risk)	+21.3	+2.81	−0.85	−2.91
With risk	+17.1	+0.92	−2.05	−3.43

The sign flips between $\mu = 3$ and $\mu = 5$ in both columns. Bisecting, the certainty-equivalent threshold is $\mu = 4.33$ against $\mu = 3.39$ with risk: risk moves the threshold by about one unit of risk aversion and does not create the effect.

Without risk the mechanism is exact. A household with J periods, no borrowing constraint and CRRA utility has $c_{t+1} = (\beta R)^{1/\mu} c_t$ and a lifetime budget, so

$$c_0 = \frac{W(R)}{D(R)}, \quad W(R) = \sum_t y_t R^{-t}, \quad D(R) = \sum_t \beta^{t/\mu} R^{t/\mu - t}$$

(`consumption_mul_pathPrice`), and saving at zero wealth rises with the rate exactly when $W(R_2)D(R_1) \leq W(R_1)D(R_2)$ (`consumption_antitone_iff`). Both terms fall with the rate,

W because future earnings are discounted harder and D because the optimal path gets cheaper, and the comparison is between the two elasticities. Differentiating,

$$\frac{d \log c_0}{d \log R} = -T_y + \left(1 - \frac{1}{\mu}\right)T_c,$$

with T_y and T_c the Macaulay durations of the two streams, so c_0 falls, and saving rises, exactly when $T_y > (1 - 1/\mu)T_c$. The predicted elasticity matches the exact one to three decimals at every μ we checked. At Aiyagari's numbers $T_y = 14.31$ and $T_c \approx 18.6$, nearly independent of μ , so the condition is $1 - 1/\mu < 0.769$, that is $\mu < 4.33$.

The economics is then plain, and has nothing to do with risk. A higher interest rate discounts future earnings more heavily, which makes the household poorer and pushes consumption down; it also makes a rising consumption path cheaper, which pushes consumption today down only to the extent that the household is willing to tilt its path. A more risk averse household will not tilt, so the second force weakens as μ grows while the first does not, and past $\mu \approx 4$ the household responds to a higher rate by consuming more today. The income stream simply has to be longer than the consumption weights, discounted by the strength of the tilt.

Theorem 5.2 (No risk, and risk aversion at most one). *If $\mu \leq 1$ the comparison holds term by term, whatever the earnings profile: the certainty-equivalent household always saves more at a higher rate.*

Formal statement: *CertaintyEquivalent.humanWealth_mul_pathPrice_le_of_le_one, CertaintyEquivalent.consumption_antitone_of_le_one.*

For $\mu > 1$ it is the duration comparison, which is a finite computation in the primitives. So a precautionary argument near zero wealth would be aimed at the wrong target: what has to be controlled is the elasticity of human wealth against the elasticity of the price of the consumption path, and risk enters only as a correction to it.

5.6 The criterion in elasticity units, and the size of the risk correction

The certainty-equivalent criterion is cleanest when written in the intertemporal elasticity of substitution rather than in risk aversion. Since it is linear in $1/\mu$,

$$T_y > \left(1 - \frac{1}{\mu}\right)T_c \quad \Longleftrightarrow \quad \frac{1}{\mu} > 1 - \frac{T_y}{T_c},$$

so the elasticity of substitution must exceed one minus the ratio of the two durations. At Aiyagari's numbers that right-hand side is 0.2306, giving $\mu < 4.33$ as before, and the identity holds to four decimals against the bisected threshold.

With risk the threshold moves down, and the move is what has to be bounded. Write

$$\Delta := \frac{1}{\mu^{\text{risky}}} - \frac{1}{\mu^{\text{CE}}}$$

for the gap in elasticity units and $s = \sigma\sqrt{1 - \rho^2}$ for the conditional standard deviation of

log earnings, the dispersion the household actually faces one period ahead. Measuring both thresholds at the same state over fifteen calibrations, $\rho \in \{0, 0.6, 0.9\}$ and σ from 0.025 to 0.4 (`numerics/gapscale.m`, `gapfit.m`):

$\rho = 0.9, \sigma$	0.025	0.05	0.1	0.2	0.4	fitted exponent
Δ	0.0034	0.0070	0.0144	0.0300	0.0612	
Δ/s	0.315	0.320	0.330	0.344	0.351	1.04

The gap is first order in the conditional standard deviation, not second order: fitting $\Delta \sim s^b$ gives $b = 1.04$ at $\rho = 0.9$, 0.89 at $\rho = 0.6$ and 0.93 at $\rho = 0$, and the ratio Δ/s stays between 0.196 and 0.351 across every calibration we ran. That first-order behaviour is itself informative: a precautionary effect on saving would be second order in the dispersion, whereas the probability that the borrowing constraint binds in a low-earnings state next period moves at first order, and that is our reading of where the correction comes from.

Taking the largest ratio observed as the bound, $\Delta \leq 0.36 s$, gives a criterion entirely in the primitives:

$$\frac{1}{\mu} \geq 1 - \frac{T_y}{T_c} + 0.36 \sigma \sqrt{1 - \rho^2}. \quad (2)$$

At Aiyagari's calibration the right-hand side is $0.2306 + 0.0628 = 0.2934$, that is $\mu \leq 3.41$, against the 3.39 that bisection on the full life-cycle model gives. The bound therefore reproduces the measured threshold to within two per cent, and it says the same thing the earlier sections said: the calibration he used for $\mu = 3$ is inside, and $\mu = 5$ is not.

Two caveats. The first inequality of (2) is a theorem about the certainty-equivalent household (`consumption_antitone_iff`, and Theorem 5.2 for $\mu \leq 1$); the constant 0.36 is measured over fifteen calibrations and is not proved. And both are statements at zero wealth in the highest earnings state, which is where the life-cycle model binds; the reduction of Theorem 5.1 is what would carry them to the rest of the state space, and its own condition fails, so this is a sharp description of where the truth lies rather than a proof of it.

5.7 The channel is the borrowing constraint

The first-order scaling points at the constraint rather than at precaution, and the point can be tested directly: loosen the borrowing limit and watch the threshold. If the constraint is the channel, the risky threshold should walk back to the certainty-equivalent one as the limit approaches the natural one, $y_{\min}/r$. At Aiyagari's calibration it does (`numerics/borrow.m`), and it does so almost linearly:

Borrowing limit	0	0.5	1	2	3	4	5	6
Threshold μ	3.42	3.50	3.58	3.74	3.89	4.05	4.21	4.38
Share of the gap closed	0%	11%	21%	40%	57%	74%	90%	104%

The natural limit here is $y_{\min}/r = 6.75$, and by a limit of 6 the gap is gone: the risky household's threshold, 4.38, is the certainty-equivalent household's, 4.33. So at this calibration the borrowing constraint accounts for the whole of the correction, and precaution accounts

for none of it. Precaution is certainly present — at the loose limit the risky household still saves 1.62 against the certainty-equivalent 1.46, a precautionary excess of eleven per cent — but it moves the level of saving without moving the sign of its response to the interest rate. Across the other calibrations we ran (`numerics/borrow3.m`) loosening the limit always closes the gap and sometimes overshoots it, so the constraint is the dominant channel rather than an exact decomposition: with ample borrowing, risk can make Theorem 1 easier than certainty equivalence rather than harder.

For the programme this is the useful part. The missing ingredient for $\mu = 3$ is not a precautionary bound but a bound on what the borrowing constraint does to the comparative static, and that is a more tractable object: without a binding constraint the household's Euler equation holds with equality at every state and Theorem 1 at zero wealth reduces to the duration criterion, which is proved outright for $\mu \leq 1$ and is an explicit finite computation in the primitives otherwise. What remains is the constrained states, which at this calibration are about eight per cent of the state space at every age.

6 The programme

The stages, in the order we intend to take them. The first is done (Section 2).

Equilibrium. A stationary overlapping-generations equilibrium: the distribution μ_1 of the youngest cohort is given (zero wealth, earnings drawn from the invariant law), μ_{j+1} is the pushforward of μ_j under the age- j policy and the earnings chain, capital supply is $K(r) = \sum_j \omega_j \int a d\mu_j$ with cohort weights ω_j , labour is exogenous, and the firm is Aiyagari's. Targets: continuity of K in the rate, a finite composition of continuous pushforwards; existence by the intermediate value theorem against the firm's demand, with the floor at the top of the interval from explicit accumulation over the life rather than from an ergodic identity; Light's Theorem 2 by induction on age, an ordering of policies at every age giving first-order stochastic dominance of every μ_j and so of K ; and single crossing against a decreasing demand.

Theorem 1 by backward induction. The research core: saving rises with the rate at every age, for $\mu \in \{3, 5\}$. The reduction of the companion write-up applies stage by stage, Theorem 1 at stage $k + 1$ following from the slack elasticity condition on the stage- k consumption functions at the pivot; the transfer lemma propagates an affine bound on the response of consumption to the rate backward from the terminal condition $c_0 = m$, where the response is exactly $\Delta r \cdot a$; and the slack condition at each stage is a finite check on the constants so obtained. The obstruction of the infinite horizon, a step multiplier of at least $1/p$, does not bite in J steps. The first session is numerical: a life-cycle economy with $J = 60$, Aiyagari's earnings chain and $\mu \in \{3, 5\}$, checking the slack condition at every age and wealth and running the recursion for the constants. If it holds the argument is formalised; if it fails only at moderate ages and wealth we look for the age-dependent refinement; if it fails badly,

log utility is the fallback, where Theorem 1 at each stage follows from the rescaling proof of the companion, which is already an induction on Bellman iterates.

Diamond (1965) and Samuelson. The two-period economy without risk: a unique steady state and monotone global convergence for logarithmic utility and Cobb–Douglas production, existence under Inada conditions and a multiplicity witness in general, dynamic inefficiency of a steady state with $r < n$ by an explicit Pareto-improving transfer, the golden rule, and the autarkic and monetary steady states of the pure-exchange economy. These are standalone results and will be the classical sections of this write-up.

Extensions. An age profile of earnings, retirement with a pension and a government budget, survival risk with accidental bequests as a second fixed point, population growth, and Lasry–Lions with age as time, whose finite-horizon core is already formalised and whose monotonicity is again the elasticity condition.

References

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- DIAMOND, P. A. (1965): “National Debt in a Neoclassical Growth Model,” *American Economic Review*, 55, 1126–1150.
- HUGGETT, M. (1996): “Wealth Distribution in Life-Cycle Economies,” *Journal of Monetary Economics*, 38, 469–494.

A Index of formal results

All in the namespace `LeanEconomics.IncomeFluctuation`, directory `LeanEconomics/OLG`. Every result depends only on the three standard axioms (`propext`, `Classical.choice`, `Quot.sound`) and none uses `sorry`.

Result	Lean name	Module
Stage value, policy, consumption	<code>stageValue,</code> <code>stageConsumption</code>	<code>stagePolicy,</code> OLG/LifeCycle
Last period: no saving	<code>stagePolicy_zero</code>	OLG/LifeCycle
Last period: consume cash on hand	<code>stageConsumption_zero</code>	OLG/LifeCycle
Positive consumption at every stage	<code>stageConsumption_pos</code>	OLG/LifeCycle
Saving and consumption rise with wealth	<code>stagePolicy_</code> <code>stageConsumption_mono</code>	<code>mono,</code> OLG/LifeCycle
Carroll–Kimball at every stage	<code>concaveOn_stageConsumption</code>	OLG/LifeCycle
Euler under the cap, by stage	<code>stage_euler_le</code>	OLG/LifeCycle
Euler above the floor, by stage	<code>stage_euler_ge</code>	OLG/LifeCycle
Finite-horizon propensities, human wealth	<code>stageMPC, stageHumanWealth</code>	OLG/LifeCycle

Result	Lean name	Module
Propensities in $(0, 1]$, above κ	stageMPC_pos, stageMPC_le_one, minMPC_le_stageMPC	OLG/LifeCycle
Lower Euler step	crra_stageMPC_step	OLG/LifeCycle
Upper Euler step	crra_stageUpper_step	OLG/LifeCycle
Theorem 2.1, lower	stageMPC_mul_le_stageConsumption	OLG/LifeCycle
Theorem 2.1, upper	stageConsumption_le_stageHumanWealth	OLG/LifeCycle
One-step saving bound	stagePolicy_le	OLG/LifeCycle
Retired household (lump-sum pension)	withPension	OLG/Retirement
Cash on hand rises with assets	withPension_resources_strictMono	OLG/Retirement
Human wealth is state-independent	stageHumanWealth_withPension_const	OLG/Retirement
Perpetuity bound $H_k \leq p/r$	stageHumanWealth_withPension_le	OLG/Retirement
Life-cycle value with retirement	lifeValue	OLG/Retirement
Saving, consumption rise with wealth	lifePolicy_mono, lifeConsumption_mono	OLG/Retirement
Pension schedules	lumpSum, tapered, cliff	OLG/MeansTest
Cash on hand	cash	OLG/MeansTest
Theorem 3.1	isOptimalSaving_mono	OLG/MeansTest
Monotone in assets when cash is	isOptimalSaving_mono_assets	OLG/MeansTest
Gentle taper keeps cash rising	cash_strictMono_tapered	OLG/MeansTest
A cliff reverses cash on hand	cash_lt_cash_of_cliff	OLG/MeansTest
Theorem 3.2	exists_consumption_decrease	OLG/MeansTest
Theorem 3.3, cliff	exists_saving_decrease_in_assets	OLG/MeansTest
Theorem 3.4	policyOf_mono_of_lt	OLG/MeansTest
One age's expectation operator	stageStep	OLG/ Equilibrium
Monotone class preserved	monoRegion_stageStep	OLG/ Equilibrium
Ordered policies, ordered functions	stageStep_le_stageStep	OLG/ Equilibrium
Life-cycle assets, aggregate capital	cohortAssets, olgCapital	OLG/ Equilibrium
Theorem 4.1	cohortAssets_le, olgCapital_le_of_stagePolicy_le	OLG/ Equilibrium
Proposition 4.2	le_stagePolicy	OLG/ Equilibrium
Theorem 4.3	olgEquilibriumRate_unique	OLG/ Equilibrium
Argmax at a concave continuation	argmax_eq_singleton_of_concaveSlices	OLG/Continuity
Slice commutes with the operator	sliceAt_bellman, sliceAt_augStageValue	OLG/Continuity
Augmented stage policy	augStagePolicy, augStagePolicy_eq	OLG/Continuity
Jointly continuous stage policy	continuousOn_augStagePolicy	OLG/Continuity
Cohort recursion in the augmented state	augCohortAssets, augCohortAssets_eq	OLG/Continuity
Theorem 4.4	continuousOn_olgSupply	OLG/Continuity

Result	Lean name	Module
Theorem 4.5	<code>exists_olgEquilibrium</code>	OLG/Continuity
Saving rises iff consumption responds little	<code>stagePolicy_le_iff_consumption_le</code>	OLG/ RateMonotone
Terminal stage, unconditional	<code>stagePolicy_mono_zero</code>	OLG/ RateMonotone
Condition (1)	<code>StageSlack</code>	OLG/ RateMonotone
Theorem 5.1, step	<code>stagePolicy_mono_step</code>	OLG/ RateMonotone
Theorem 5.1, induction	<code>stagePolicy_mono_of_stageSlack</code>	OLG/ RateMonotone
Theorem 5.1, aggregate	<code>olgCapital_le_of_stageSlack</code>	OLG/ RateMonotone
Human wealth, price of the path	<code>humanWealth, pathPrice</code>	OLG/Certainty Equivalent
Closed form $c_0 = W/D$	<code>consumption_mul_pathPrice</code>	OLG/Certainty Equivalent
Saving rises iff the cross product falls	<code>consumption_antitone_iff</code>	OLG/Certainty Equivalent
Theorem 5.2	<code>consumption_antitone_of_le_one</code>	OLG/Certainty Equivalent